

VID (Virtual Identification) — Implementation Plan

Principle: One institution → one connected ecosystem → one student master record → multiple connected services. VID is *not* a set of disconnected CRUD modules.

Lifecycle VID must manage



Platform must support: multi-institution architecture · role-based access · modular workspaces · API-first · responsive web app · Parent/Student mobile app · AI services via AI Yantra · optional institution-specific modules.

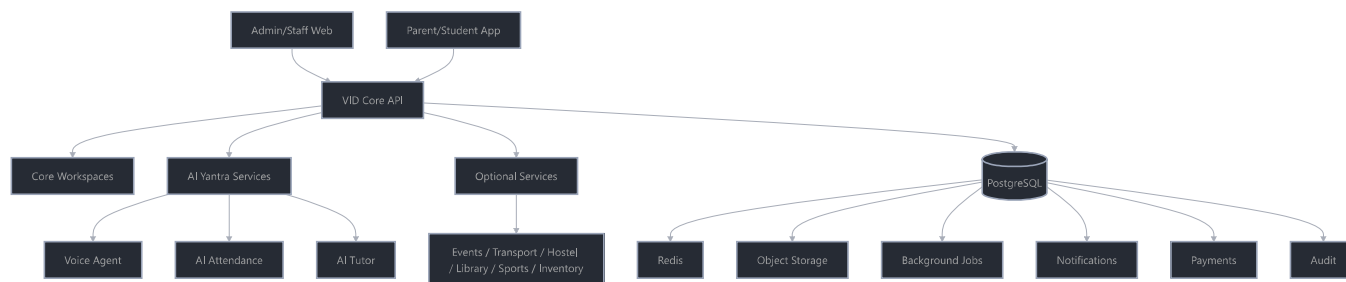
1. Workspace Classification (final)

#	Core workspace	Interface	Phase
1	Super Admin	Web	1
2	Institution Admin	Web	1
3	Admissions	Web	1
4	Academics	Web	1
5	Faculty	Web	1
6	Attendance	Web	2
7	Examinations	Web	2
8	Finance & Fee Management	Web + App	2
9	Documents	Web + App	2
10	Staff Management / HRMS	Web	2
11	Timetable	Web + App	2
12	Yantra Voice Agent	Web	3
13	Yantra AI Attendance	Web	3
14	Yantra AI Tutor	Web + App	3
#	Optional workspace	Interface	Phase
1	Events	Web + App	4
2	Transport Management	Web + App	4

3	Hostel Management	Web + App	4
4	Library Management	Web + App	4
5	Sports Management	Web + App	4
6	Inventory & Asset Management	Web	4

Students are NOT a workspace — a central entity used everywhere. Optional modules: independently enabled/disabled per institution, hidden from navigation when disabled, isolated from core, connected to shared Student/User entities, permission-gated, never mandatory for the core academic engine.

2. Platform Architecture



Frontend, mobile, backend, AI services, database and shared services communicate only through clearly defined interfaces.

3. Multi-Tenant Architecture

VID SaaS → Institution A / B / C, each with Admin, Staff, Faculty, Students, Parents. **Every tenant-owned record carries institution_id** (directly or via a securely enforced parent). Isolation enforced at: **API · Authorization · Database query · File storage · AI context**. No user may access another institution's data.

4. Roles, RBAC & Authorization

Role	Scope
Super Admin	All institutions
Institution Admin	One institution
Admission Team	Admissions
Academic Coordinator	Academic data
Faculty	Assigned classes/subjects
Exam Team	Examinations
Finance Team	Finance
HR / Staff	HRMS
Student	Own data

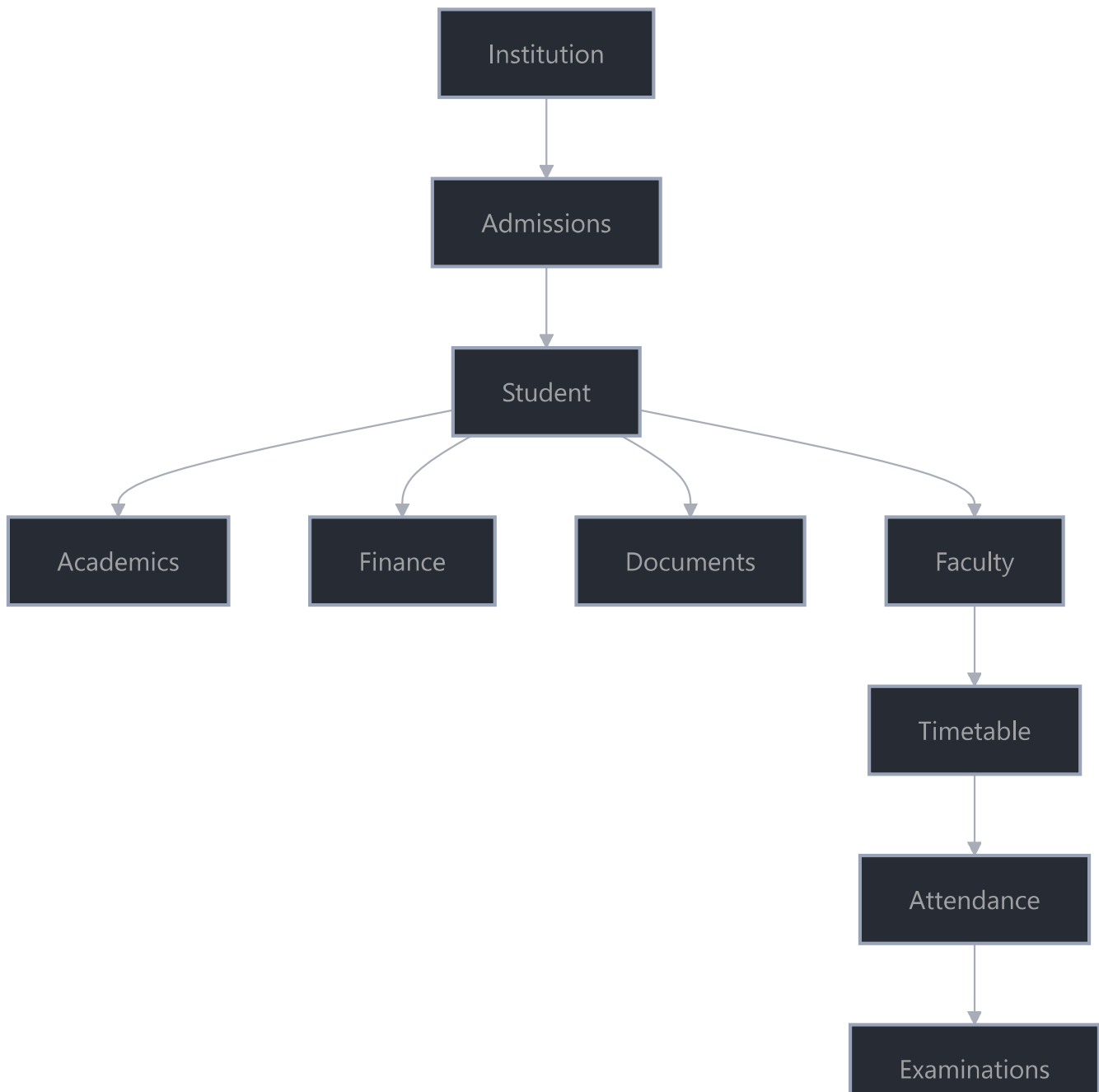
Parent

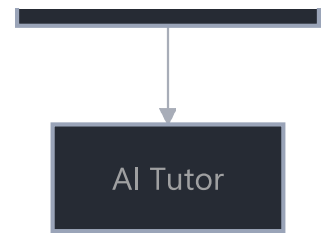
Own children only (rule 9)



Example: Faculty → attendance.create → Class 8-A → Mathematics → authorized students. Faculty must not automatically see unrelated classes, finance or institution-wide data.

5. Delivery Roadmap & Dependencies





Optional modules attach to Student and Institution only.

6. Super Admin & Institution Admin (Phase 1)

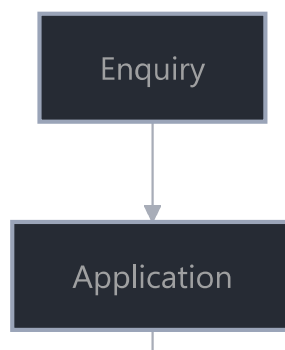
Super Admin pages

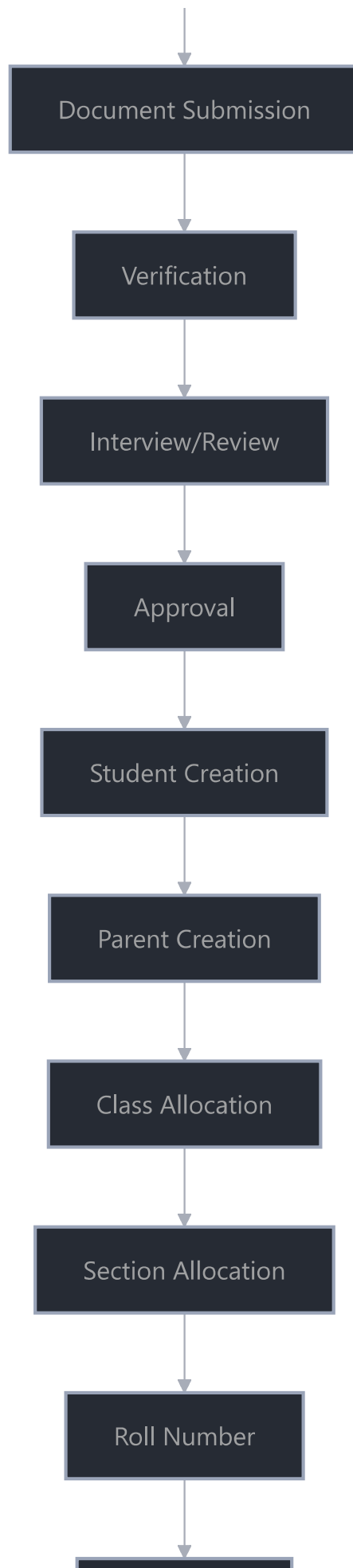
Area	Pages
Dashboard	Dashboard
Institutions	Institutions, Institution Onboarding
Plans	Institution Plans, Subscriptions
Users	Institution Users
Security	Global Roles, Global Permissions
Monitoring	Platform Audit Logs, System Health
AI	AI Yantra Configuration
Billing	Billing
Settings	Platform Settings

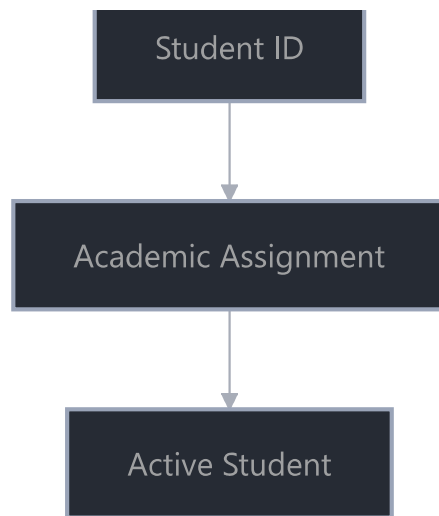
To do: create/activate/deactivate institutions · create institution administrators · manage subscriptions · configure global permissions and global AI services · monitor system health · view platform audit.

Institution Admin must manage: Institution Profile, Academic Years, Departments, Courses, Classes, Sections, Subjects, Users, Roles, Permissions, Admissions, Academics, Faculty, Attendance, Examinations, Finance, Documents, HRMS, Timetable, AI Yantra, enabled optional modules.

7. Admissions & Student Lifecycle (Phase 1)



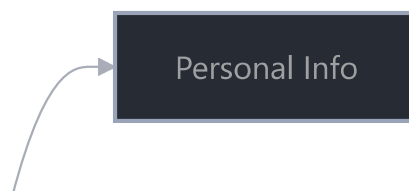


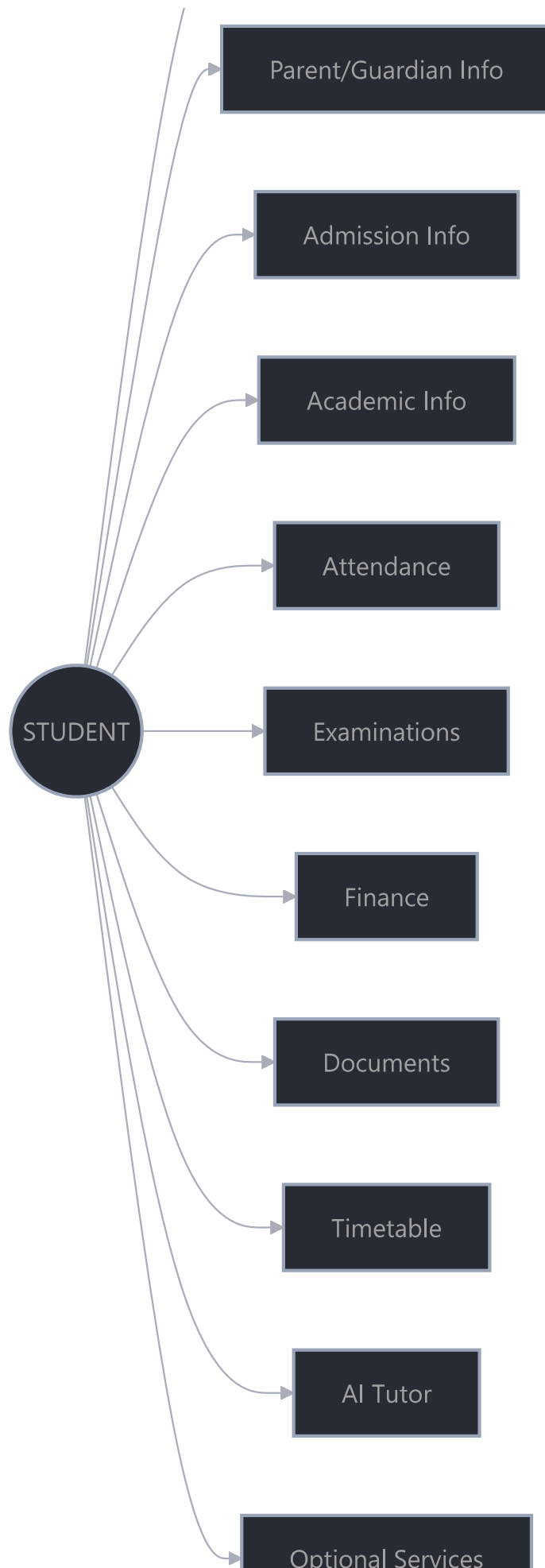


Category	Pages / what to build
Dashboard	Admissions Dashboard
Enquiries	Enquiries
Applications	Applications, Applicants
Verification	Application Review, Document Verification
Decision	Admission Approval, Rejected, Waitlisted
Registration	Student Registration
Student	Student Profile
Parent	Parent / Guardian Details
Allocation	Class, Section, Academic Year
Identification	Roll Number, Student ID
Documents	Admission Documents
Finance	Admission Fee
History	Admission History

Entities: admissions, applications, admission_documents → on approval create students , parents , guardians , student_parents . **Notifications:** Applicant/Parent. **API:** /api/v1/admissions .

8. Student Master Entity







All modules reference the **same** student record — no duplicates, no module-specific student records. **API:** `/api/v1/students` .

9. Academics (Phase 1)



Category	Pages
Dashboard	Academic Dashboard
Structure	Academic Years, Departments, Courses, Classes, Sections
Subjects	Subjects
Curriculum	Curriculum, Syllabus
Calendar	Academic Calendar, Academic Sessions
Assignment	Faculty Assignment, Class Assignment, Subject Assignment
Students	Student Academic Records
Progress	Academic Progress
Promotion	Promotion / Demotion
Configuration	Academic Configuration

Entities: academic_years, departments, courses, classes, sections, subjects, curriculum, syllabus.

10. Faculty (Phase 1)

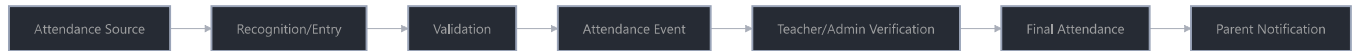
Area	Pages
Dashboard	Faculty Dashboard
Profile	My Profile
Teaching	My Classes, My Sections, My Subjects
Students	My Students, Student Academic Profile
Schedule	Today's Timetable
Attendance	Attendance
Teaching	Assignments, Study Materials
Assessment	Marks, Examinations

AI	AI Tutor
Performance	Student Performance
HR	Leave, Workload

Permissions scoped to assigned: Classes, Sections, Subjects, Students, Timetable. **Entities:** faculty, faculty_subjects, faculty_classes. **Journey:** Login → Dashboard → Today's Timetable → Class → Attendance → Teaching → Assignments/Materials → Marks → Student Performance → AI Tutor Support.

11. Attendance (Phase 2, mandatory core)

Sources: AI Face (Yantra AI Attendance), CCTV (camera recognition).



Pages: Attendance Dashboard, Daily, Class, Section, Subject, Faculty Attendance Entry, Student Attendance, History, Absentees, Late Arrivals, Leave Records, Corrections, Approval, Calendar, Configuration, Logs. **Entities:** attendance, attendance_records, attendance_sessions, attendance_corrections. **Rule:** AI-generated attendance must pass validation before becoming final.

12. Examinations (Phase 2)

Category	Pages
Dashboard	Exam Dashboard
Configuration	Exam Types, Exam Configuration
Schedule	Exam Schedule, Exam Subjects
Rooms	Exam Rooms
Allocation	Student Allocation, Invigilators
Papers	Question Papers
Marks	Marks Entry, Marks Import
Verification	Marks Verification, Marks Approval
Results	Grades, Results
Reports	Report Cards, Student Result History
Notifications	Exam Notifications
Import	Excel Import



Invalid data must never be silently committed. **Entities:** exams, exam_types, exam_schedules, exam_subjects,

marks, grades, report_cards.

13. Finance & Fee Management (Phase 2)

Category	Pages
Dashboard	Finance Dashboard
Configuration	Fee Structure, Categories, Groups
Assignment	Student Fees, Fee Assignment
Billing	Invoices, Pending Fees
Collection	Fee Collection, Payments
Online	Online Payments, Payment Gateway
Adjustments	Discounts, Scholarships
Refunds	Refunds
Receipts	Receipts
Reconciliation	Payment Reconciliation
Reports	Financial Reports
Audit	Finance Audit Logs



Transactions auditable; historical records never simply overwritten. **Entities:** fee_structures, student_fees, invoices, payments, transactions, receipts, discounts, scholarships, refunds.

14. Documents & HRMS (Phase 2)

Documents pages: Dashboard, Student Documents, Staff Documents, Institution Documents, Admission Documents, Certificates, Bonafide Certificates, ID Documents, Circulars, Templates, Document Verification, Document Requests, Document History. **Formats:** PDF, JPG, PNG, DOC, DOCX. **Entities:** documents, document_templates.

HRMS category	Pages
Dashboard	HRMS Dashboard
Staff	Staff Directory, Staff Profiles
Organization	Departments, Designations
Employment	Employment Details
Documents	Staff Documents

Attendance	Staff Attendance
Leave	Leave Management, Requests, Approval
Workload	Workload, Faculty Workload
History	Staff History
Payroll	Payroll Integration (may start as external boundary)
Reports	HR Reports

Entities: staff, departments, designations, staff_attendance, leave_requests.

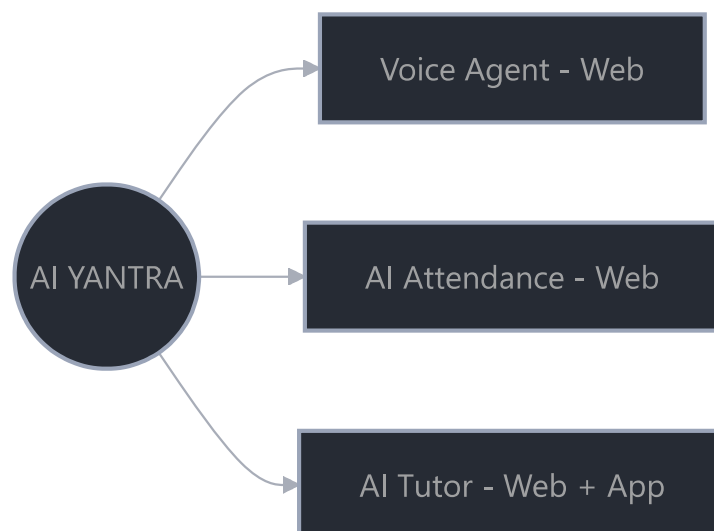
15. Timetable (Phase 2, core)

Pages: Dashboard, Academic, Class, Section, Faculty, Subject, Room Timetable, Period Configuration, Working Days, Breaks, Substitution, Free Periods, Timetable Conflicts, Timetable Publishing, Conflict Detection.

Conflict	Detect
Teacher	Teacher assigned twice
Room	Room assigned twice
Class	Conflicting subjects
Section	Conflicting periods
Period	Invalid period allocation

Entities: timetables, periods, rooms.

16. AI Yantra (Phase 3) — exactly three workspaces



No additional AI Yantra workspaces unless the product definition changes. AI services never get unrestricted DB access.

16.1 Voice Agent

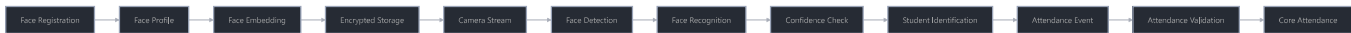
Pages: Dashboard, Campaigns, Create Campaign, Call Recipients, Call Templates, Configuration, Languages, Voice Configuration, Call Queue, Calls (Active/Completed/Failed), Call History, Interaction History, Scheduled Calls, Voice Agent Logs. **Use cases:** Fee Reminder, Attendance Alert, Exam Reminder, PTM Reminder, Event Announcement, Admission Follow-up, Emergency Announcement, General Announcement.



Entities: ai_voice_campaigns, ai_voice_calls. **Tech:** STT + TTS, LLM API.

16.2 AI Attendance

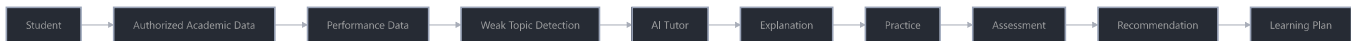
Pages: Dashboard, Face Registration, Student Face Profiles, Face Recognition, Camera Management, CCTV Devices, Classroom Cameras, Live Recognition, Attendance Verification, Recognition Logs, Unknown Faces, Attendance Corrections, Device Health, Configuration.



Biometric data needs strong security, access restrictions and retention policies. **Entities:** face_profiles, face_embeddings, ai_attendance_events. **Tech:** OpenCV + face recognition model.

16.3 AI Tutor

Pages: Dashboard, My Subjects, Ask Tutor, Topic Explanation, Doubt Resolution, Practice Questions, Question Generator, Mock Exams, Weak Topic Detection, Performance Analysis, Learning Recommendations, Personal Study Plan, Learning History, Progress Tracking.



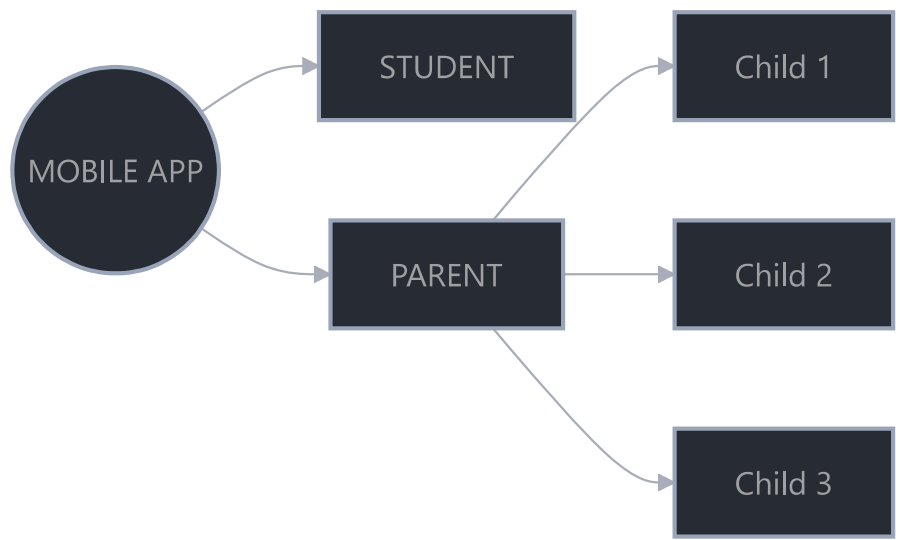
Only authorized student/context data. **Entities:** tutor_sessions, tutor_messages, learning_profiles, recommendations. **Tech:** LLM API, vector DB where required.

17. Optional Modules (Phase 4)

Module	Capabilities
Events	Events, registrations, participants, attendance, certificates
Transport	Routes, stops, vehicles, drivers, tracking, allocation
Hostel	Hostels, rooms, beds, allocation, attendance, fees
Library	Books, copies, issue, return, fines, reservations
Sports	Teams, coaches, training, competitions, results
Inventory	Assets, stock, vendors, purchases, allocation, maintenance

Requirements: independently toggled · hidden when disabled · isolated from core · linked to shared Student/User · permission-gated.

18. Mobile Application (one app, role-based)



Student bottom nav: Home (Overview) · Academics (classes, subjects, materials) · Attendance · Fees (Finance) · Profile (personal info). **Student Home shows:** Welcome, Today's Timetable, Today's Attendance, Upcoming Exams, Pending Fees, Assignments, AI Tutor, Announcements. **Additional sections:** Academics, Attendance, Examinations, Finance, AI Tutor, Documents, Timetable. **Parent:** multiple children from one login; Child Profile, Attendance, Academics, Exams, Fees, Timetable, Documents, Notifications, AI Tutor where permitted. Parent journey: Login → Select Child → Dashboard.

19. Responsive Web Design

Widths: 320+ small mobile · 375+ mobile · 390+ / 414+ large mobile · 768+ tablet · 1024+ small desktop · 1280+ desktop · 1440+ large desktop.

Device	Layout
Desktop	Sidebar + Topbar + Content
Tablet	Collapsible sidebar, adaptive width, responsive cards/forms/dialogs/charts
Mobile	☰ VID 🔔 header, content, bottom bar Home / Search / Profile; not a scaled desktop

Tables: desktop sorting, filtering, pagination, bulk actions, column visibility; mobile cards or internal scrolling; never page-level horizontal scroll. **Forms:** 320–414px 1 col · 768px 1–2 · 1024px 2 · 1280px+ 2–4; inline validation, searchable selectors, mobile date pickers, bottom sheets, sticky actions, clear error states. **Dashboards:** desktop 4 cards → 2 charts; mobile single column stacking.

20. Technical Stack & Backend Structure

Layer	Technology
Frontend	React / Next.js, TypeScript, Tailwind CSS
Data fetching / validation	TanStack Query, Zod
Database	PostgreSQL
Auth	NextAuth.js
API	Express.js
Deployment	Vercel

Backend	Python + FastAPI (Flask allowed if standardized; FastAPI preferred)
Database / Cache	PostgreSQL / Redis
Jobs	Celery + Redis or managed queue
Storage	S3-compatible object storage
AI	Python AI services, OpenCV + face model, LLM API, STT + TTS, vector DB where required

backend/: auth, tenants, users, admissions, students, academics, faculty, attendance, examinations, finance, documents, hrms, timetable, events, transport, hostel, library, sports, inventory, ai_yantra/{voice_agent, ai_attendance, ai_tutor}, notifications, audit, common.

21. Data & API Architecture

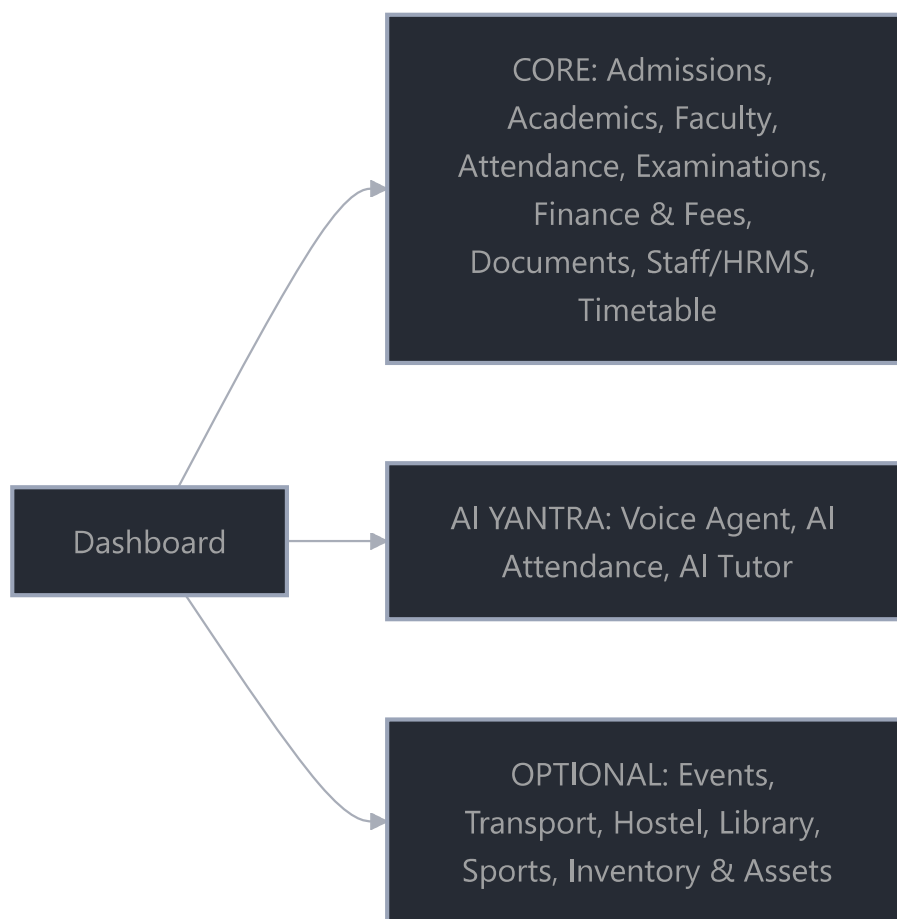
Domain	Entities
Platform	institutions
Auth	users, roles, permissions, user_roles
Admissions	admissions, applications, admission_documents
Students	students, parents, guardians, student_parents
Academics	academic_years, departments, courses, classes, sections, subjects
Curriculum	curriculum, syllabus
Faculty	faculty, faculty_subjects, faculty_classes
Attendance	attendance, attendance_records, attendance_sessions, attendance_corrections
Exams	exams, exam_types, exam_schedules, exam_subjects, marks, grades, report_cards
Finance	fee_structures, student_fees, invoices, payments, transactions, receipts, discounts, scholarships, refunds
Documents	documents, document_templates
HRMS	staff, departments, designations, staff_attendance, leave_requests
Timetable	timetables, periods, rooms
Voice / AI Attendance / AI Tutor	as in section 16

Every major tenant-owned entity includes `institution_id`. **Versioned APIs:** `/api/v1/` + auth, institutions, users, admissions, students, academics, faculty, attendance, examinations, finance, documents, hrms, timetable, events, transport, hostel, library, sports, inventory, vantra/voice, vantra/attendance, vantra/tutor.

22. Security, Audit, Files & Notifications

Security: HTTPS, secure password hashing, JWT/secure sessions, refresh token rotation, RBAC, tenant isolation, permission middleware, resource-level authorization, rate limiting, input validation, secure headers, CSRF where applicable, file validation, malware scanning, private object storage, signed URLs. **Audit fields:** Actor, Action, Resource, Old Value, New Value, Timestamp, Institution, IP/Device where appropriate (e.g. Teacher A updated attendance for Student 1023, ABSENT → PRESENT). **Notifications (shared service, no Communication workspace):** Attendance → Parent/Student · Finance → Parent/Student · Exams → Student/Parent · Admissions → Applicant/Parent · Events → Participants. **Channels:** Push, SMS, Email, Yantra Voice. **Files (object storage):** institutions/, students/, photos/, documents/, staff/, examinations/, question-papers/, report-cards/, finance/, receipts/, ai/. DB stores metadata and references only.

23. Navigation (Institution Admin)



24. Student & Finance Journeys



Parent dashboard: Attendance, Academics, Exams, Fees, Timetable, Documents, AI Tutor. Finance journey: Fee Structure → Fee Assignment → Invoice → Parent → Payment → Gateway → Verification → Transaction → Receipt → Notification.

25. Performance & Quality

metric	target
Normal page load	< 2.5 s
Standard API response	< 500 ms
Standard DB query	< 200 ms
Production availability	99.9%+ where practical

AI operations may take longer. **Quality checklist:** consistent UI components and design system; loading, empty, error, success and permission states; responsive layouts; accessibility; form validation; search/filtering; pagination; confirmation for destructive actions; proper audit trails; no broken navigation; no page-level horizontal scroll; no inaccessible optional modules; no cross-tenant data exposure.

26. The 30 Non-Negotiable Rules (acceptance checklist)

1. Student is a central master entity, not a workspace.
2. Core workspaces always available.
3. AI Yantra has exactly three workspaces.
4. Optional modules independently enabled/disabled.
5. Disabled modules vanish from navigation.
6. Every tenant-owned record institution-isolated.
7. RBAC combined with resource-level authorization.
8. Faculty access limited to assigned resources.
9. Parent access limited to their children.
10. Students access only their own permitted data.
11. AI services never get unrestricted DB access.
12. AI Attendance passes validation before final attendance.
13. Financial transactions remain auditable.
14. Important changes generate audit events.
15. Large files use object storage.
16. Notifications are a shared service, not a workspace.
17. APIs are versioned.
18. Platform is API-first.
19. Admin/staff app fully responsive.
20. Mobile app has dedicated Parent/Student experiences.
21. No page-level horizontal scrolling.
22. Tables have responsive mobile behavior.
23. Forms adapt to screen size.
24. Dashboards automatically reflow.
25. Navigation adapts by role and screen size.

25. Navigation adapts by role and screen size.

26. Security enforced server-side, not only by hiding UI.

27. Optional services never break the core platform.

28. All connected workflows use shared master entities.

29. Performance, accessibility, security and responsiveness are product requirements, not later enhancements.

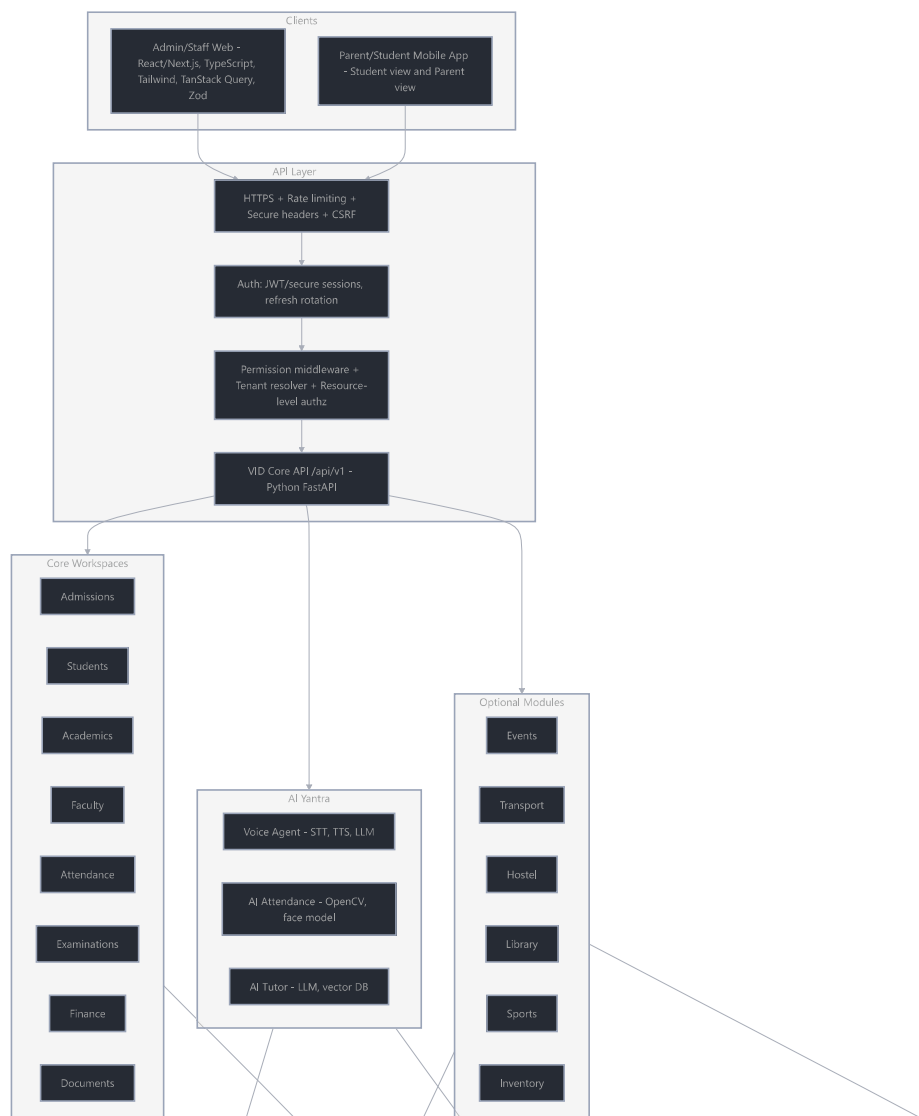
30. The final system operates as one connected VID ecosystem, not disconnected applications.

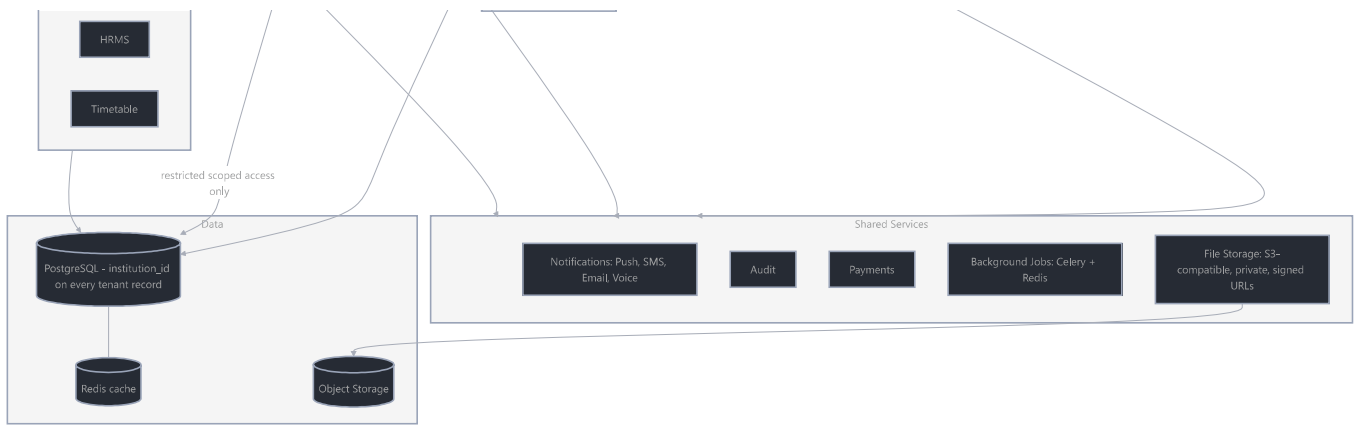
27. Final Build Instruction

Implement the complete platform per these 30 points, preserving workspace boundaries, role permissions, multi-tenant architecture, connected student lifecycle, AI Yantra structure, optional-module architecture, responsive behavior, API architecture, security requirements, database relationships and user journeys. Do **not** add workspaces, duplicate entities, disconnected module-specific student records, or desktop-only interfaces unless a future product change requires it.

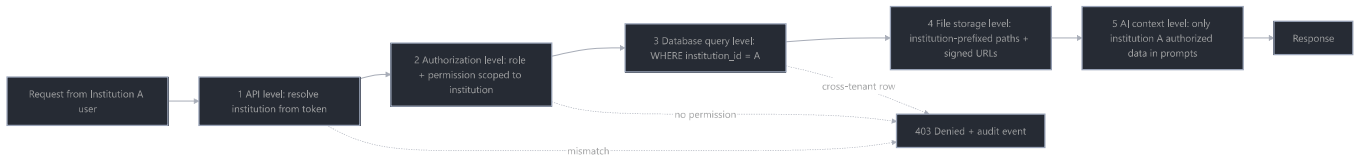
PART B — Detailed Diagrams

B1. Full System Architecture (layers and shared services)

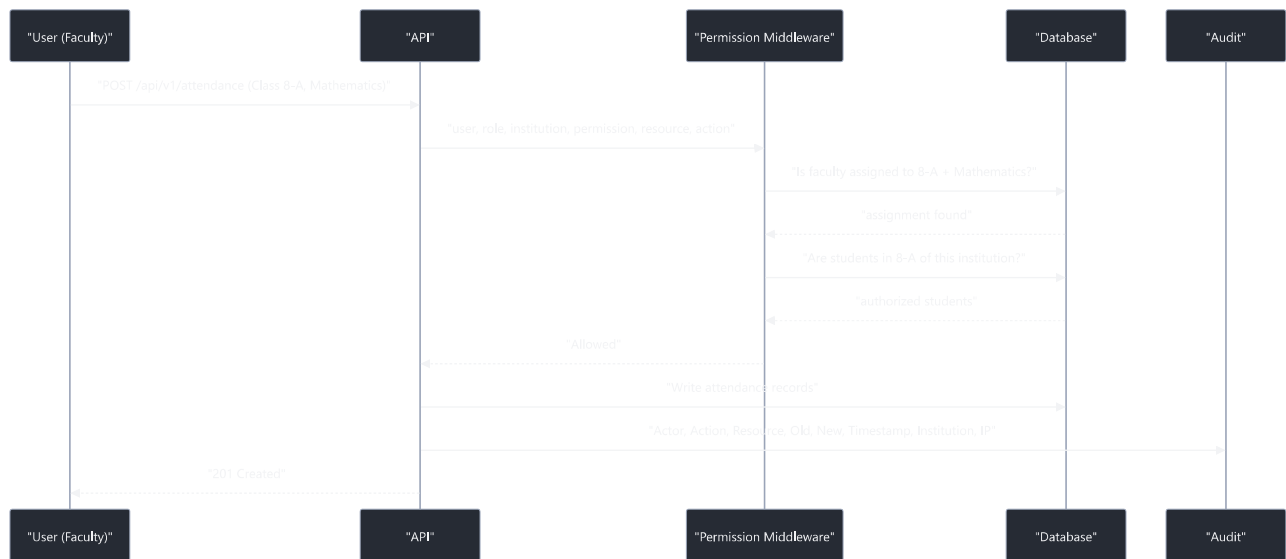




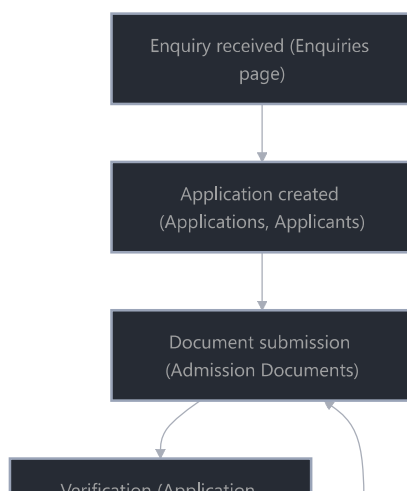
B2. Multi-Tenant Isolation (five enforcement points)

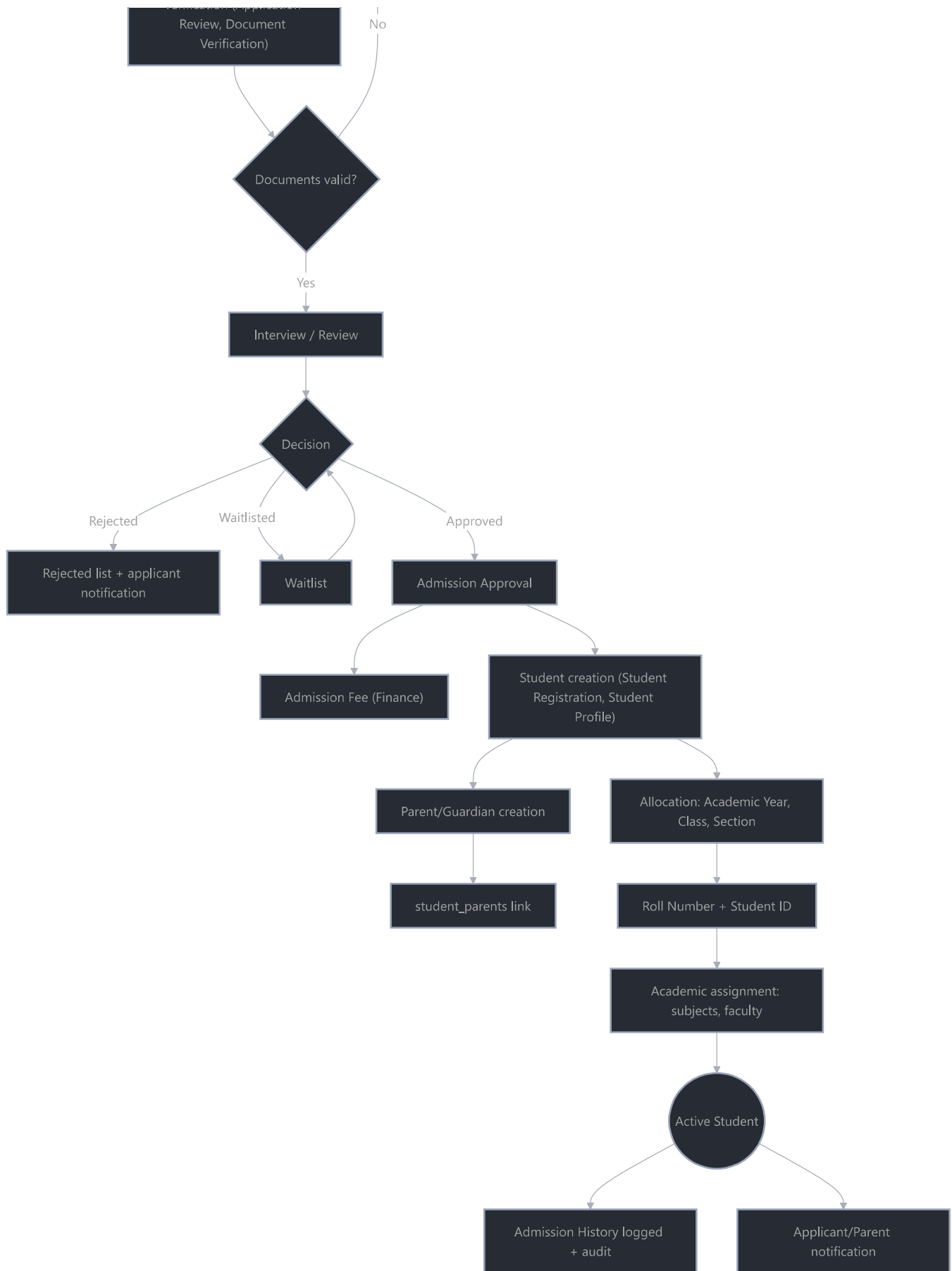


B3. Authorization Decision Sequence

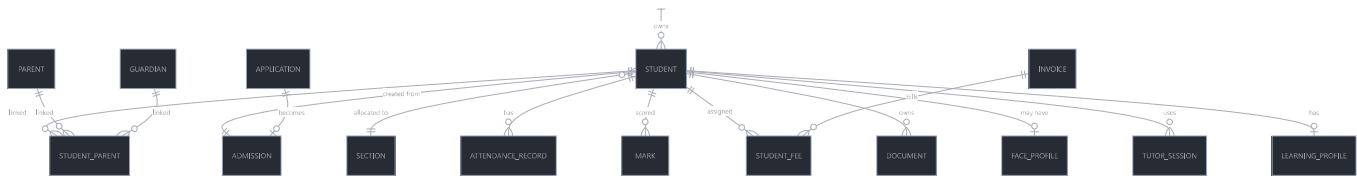


B4. Admissions — Detailed Flow with Decisions





B5. Student Master Record — Entity Relationships

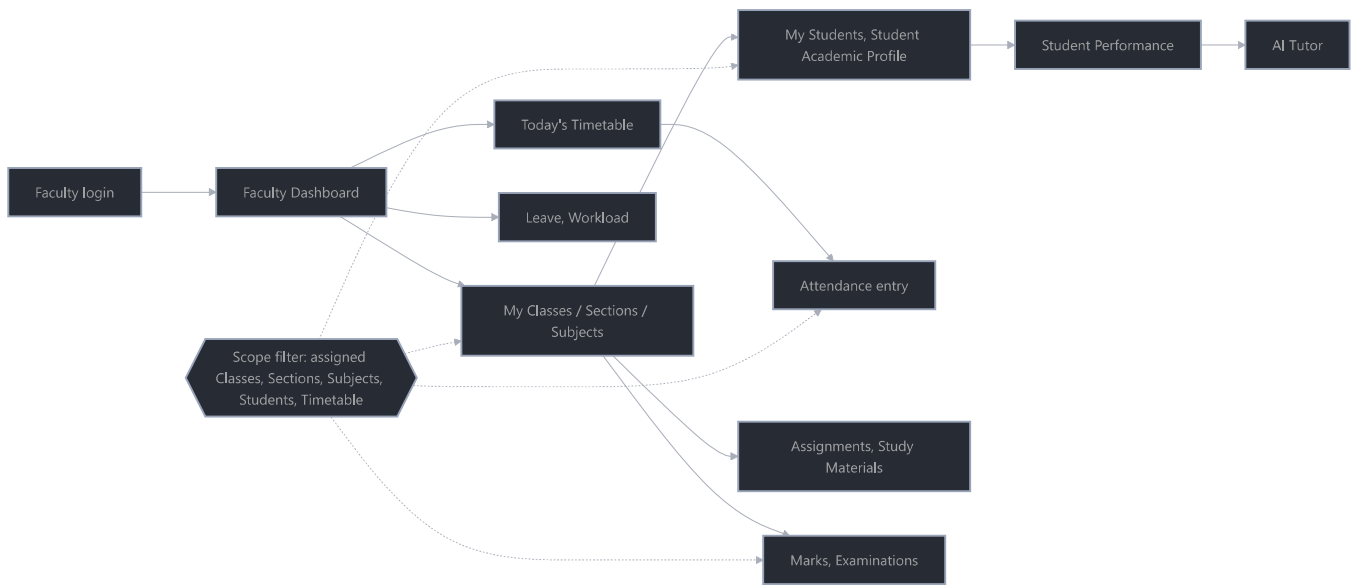


B6. Academic Hierarchy and Assignment

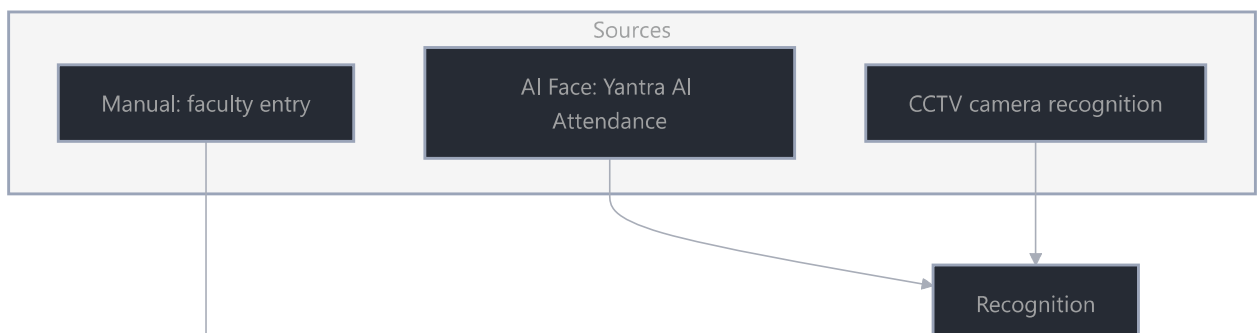
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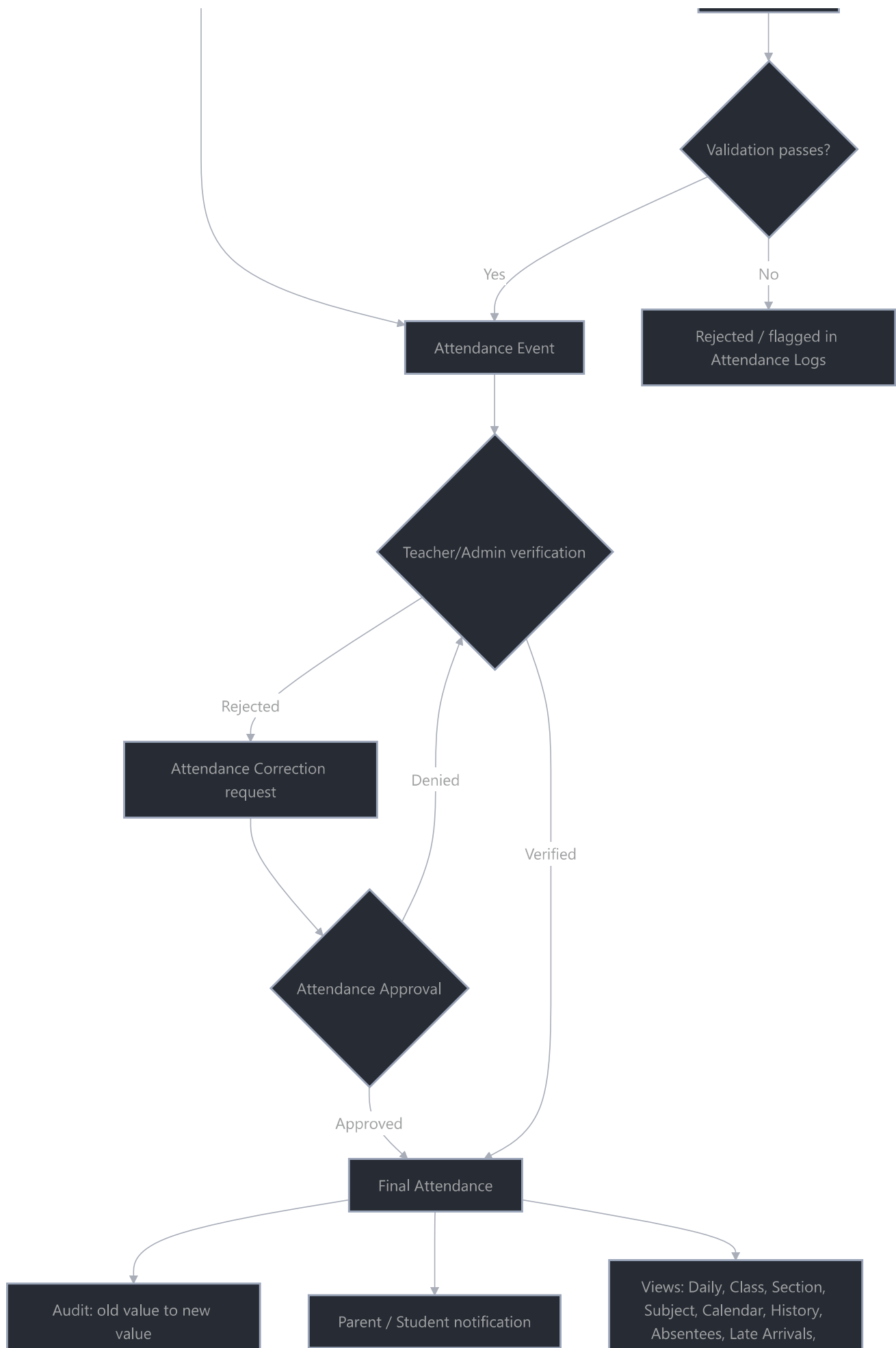
erDiagram
    INSTITUTION ||--o{ ACADEMIC_YEAR : "has"
    ACADEMIC_YEAR ||--o{ DEPARTMENT : "contains"
    DEPARTMENT ||--o{ COURSE : "offers"
    COURSE ||--o{ CLASS : "has"
    CLASS ||--o{ SECTION : "divided into"
    COURSE ||--o{ SUBJECT : "includes"
    SUBJECT ||--o{ CURRICULUM : "defined by"
    CURRICULUM ||--o{ SYLLABUS : "detailed in"
    FACULTY ||--o{ FACULTY_SUBJECT : "teaches"
    SUBJECT ||--o{ FACULTY_SUBJECT : "taught by"
    FACULTY ||--o{ FACULTY_CLASS : "assigned"
    SECTION ||--o{ FACULTY_CLASS : "assigned"
    SECTION ||--o{ STUDENT : "enrolls"
  
```

B7. Faculty Workspace — Scoped Access Map

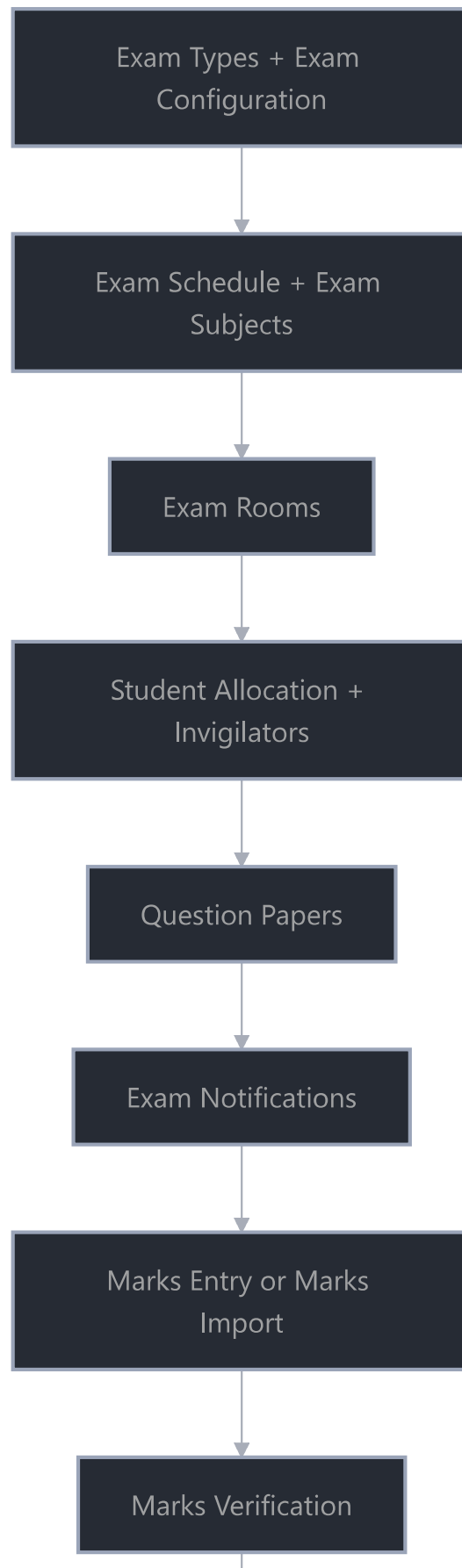


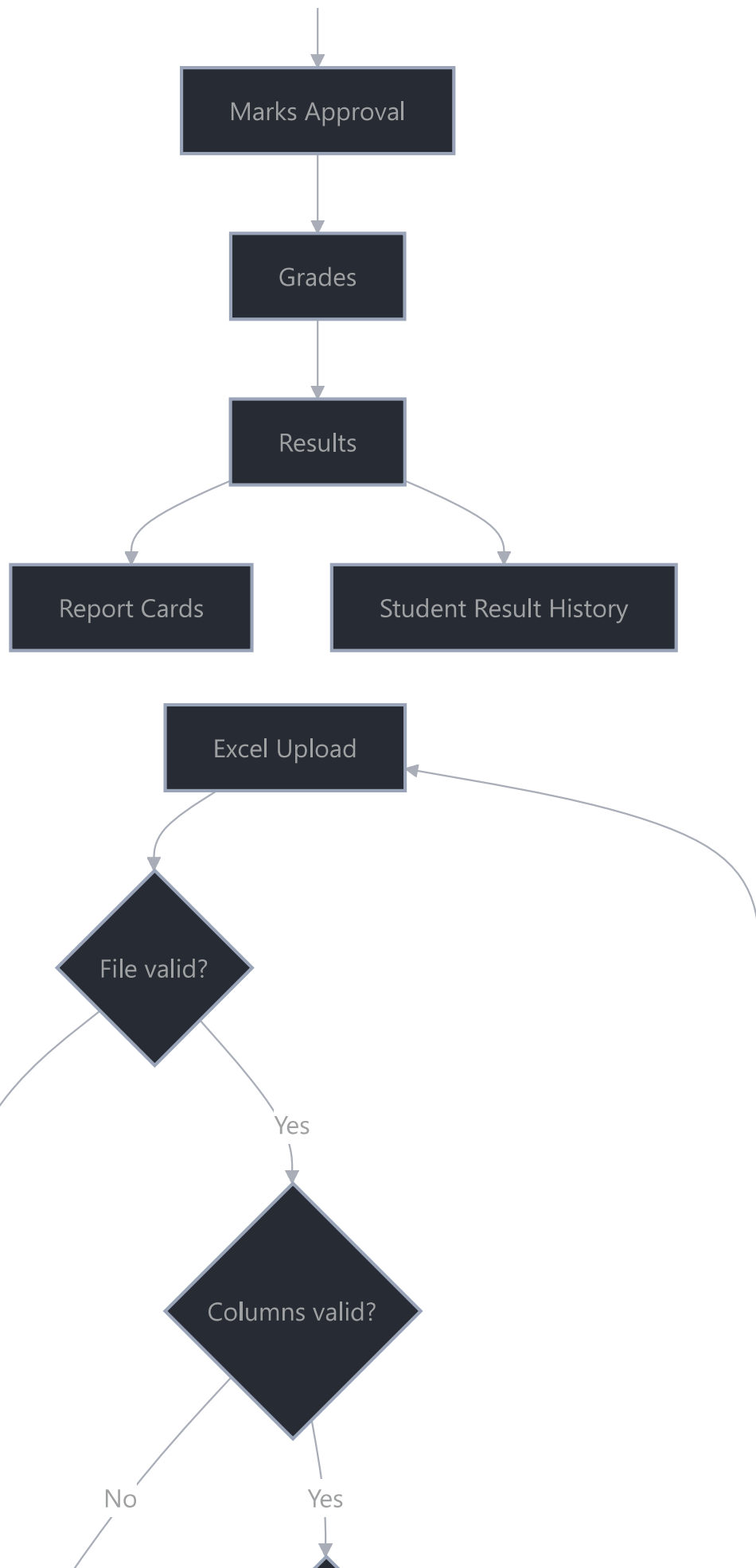
B8. Attendance — Detailed Flow with Validation and Corrections

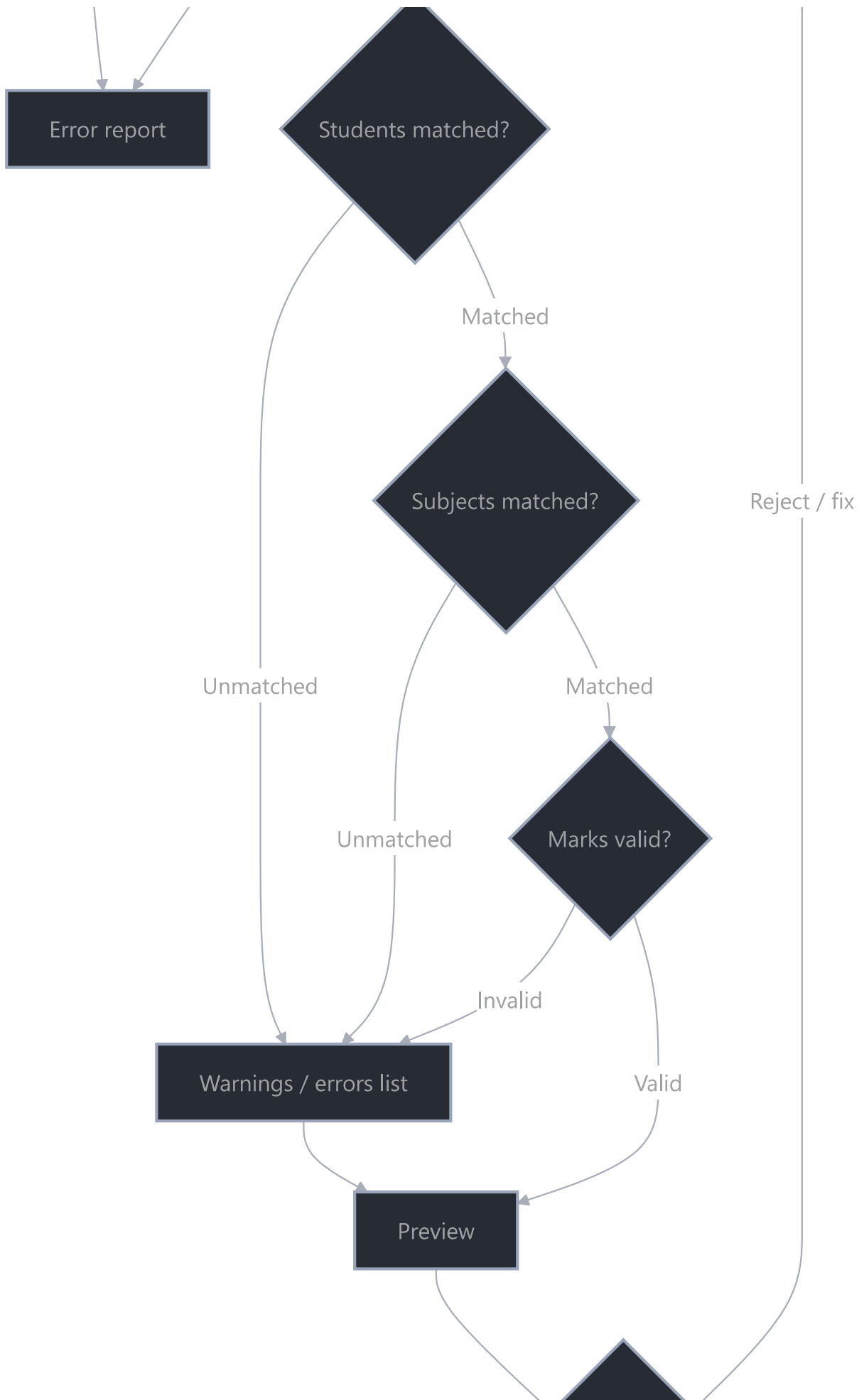


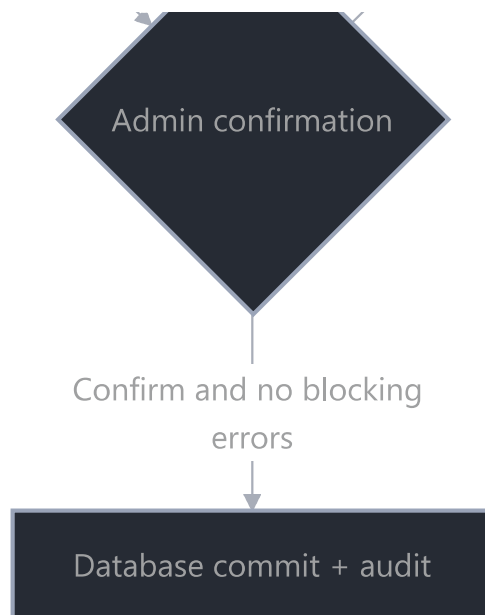


B9. Examinations — Lifecycle and Excel Import with Error Paths

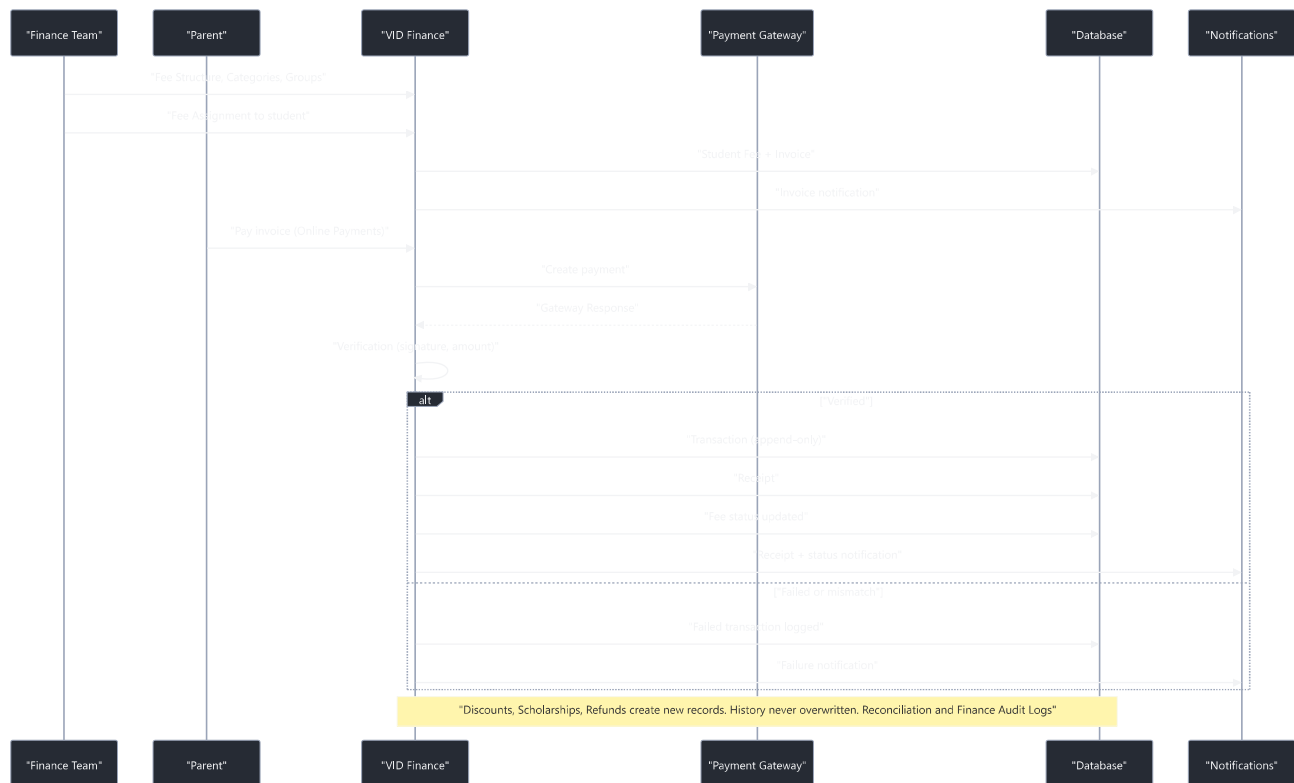




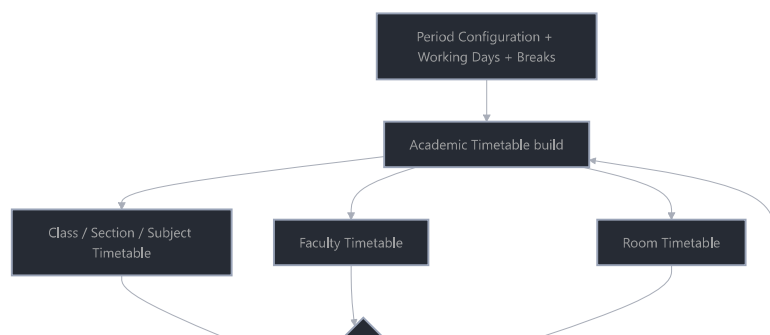


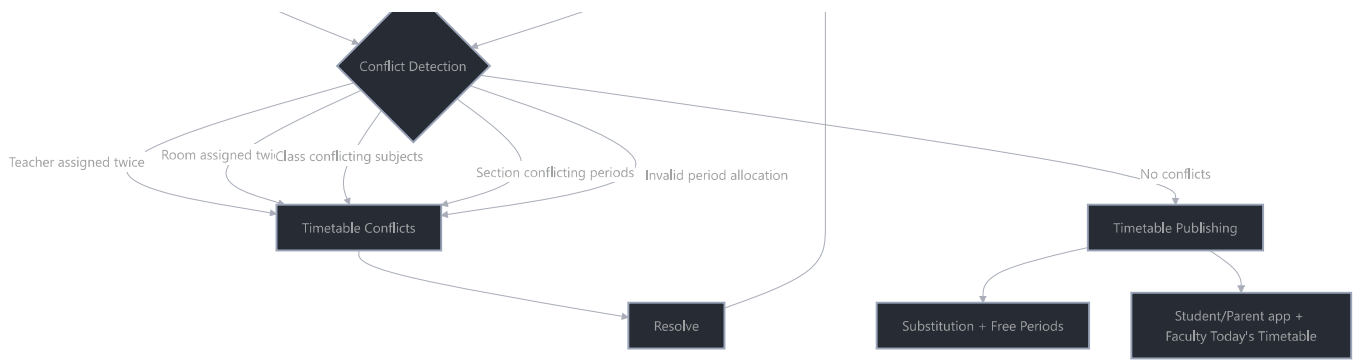


B10. Finance — Payment Sequence and Failure Handling

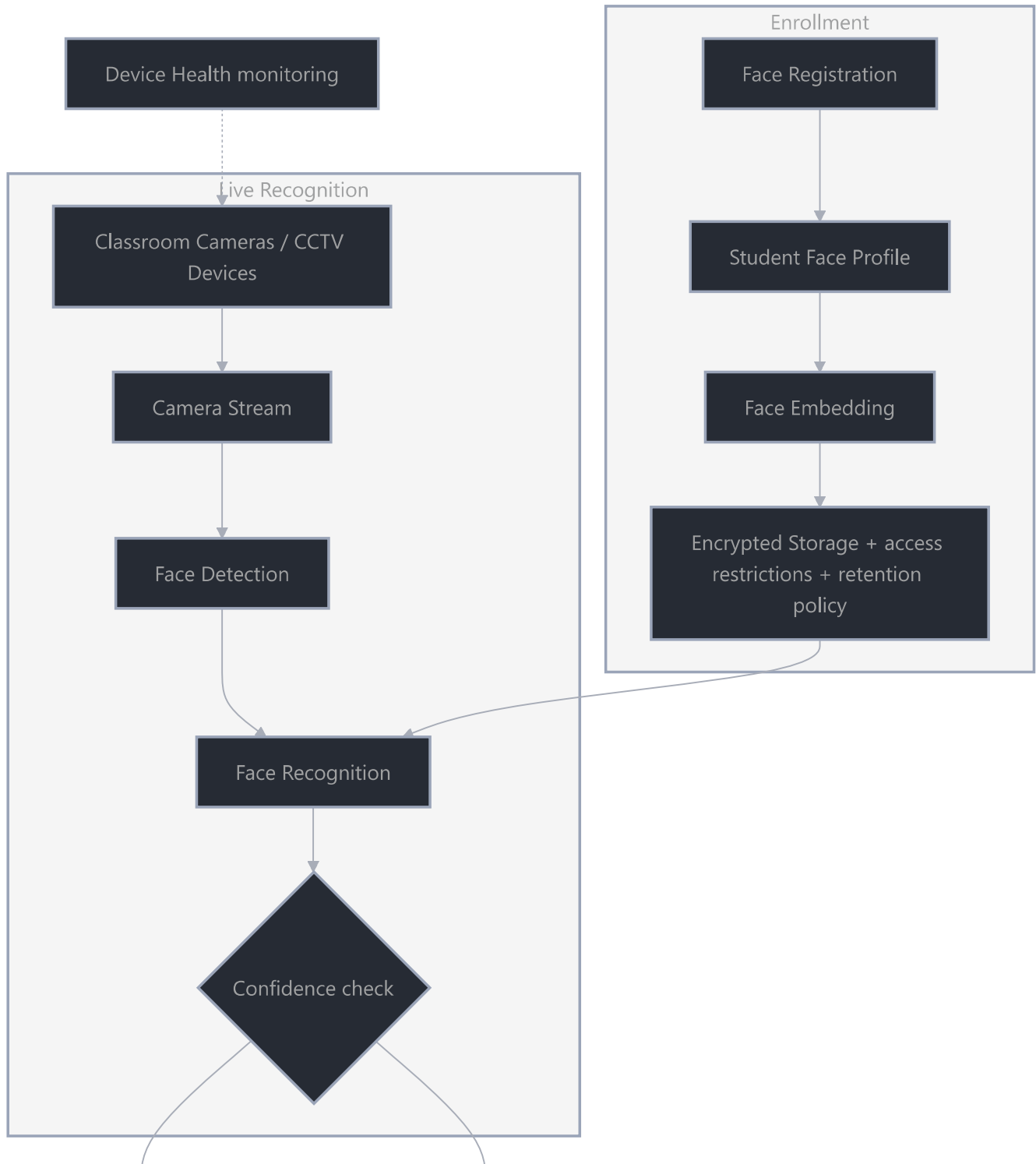


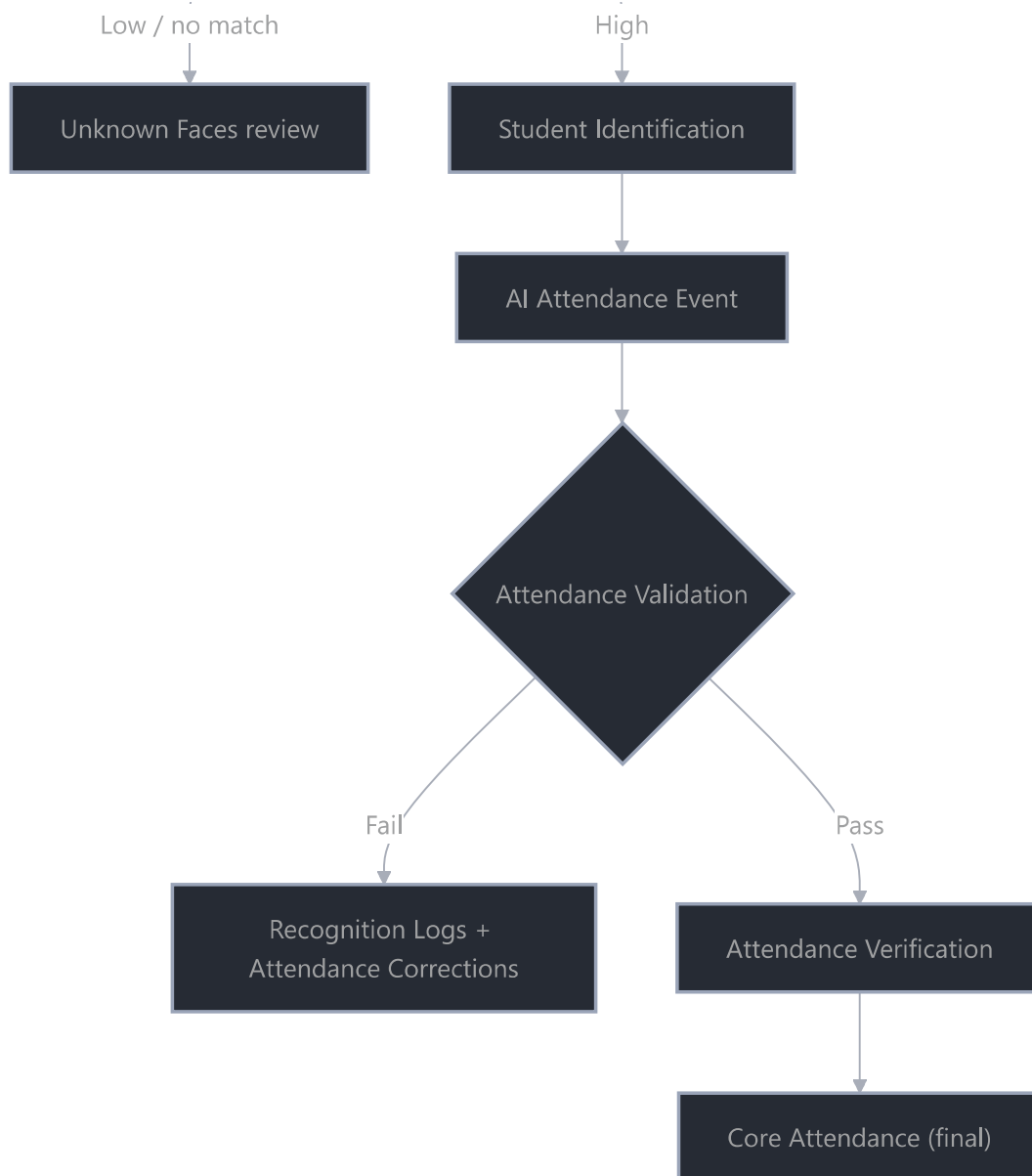
B11. Timetable — Build, Conflict Detection and Publishing



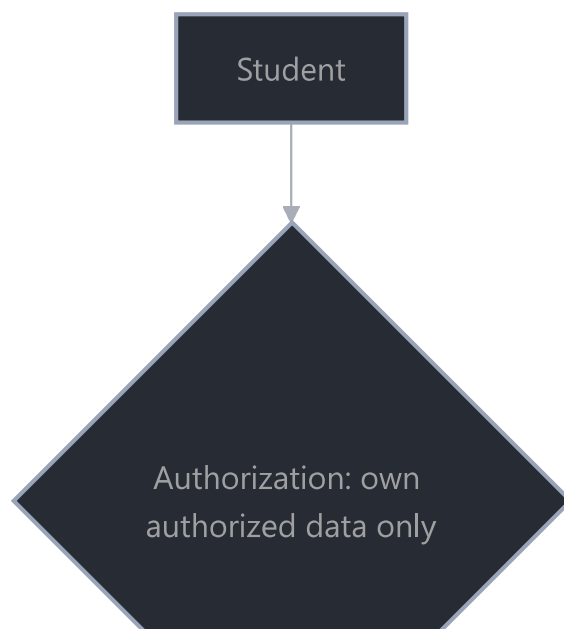


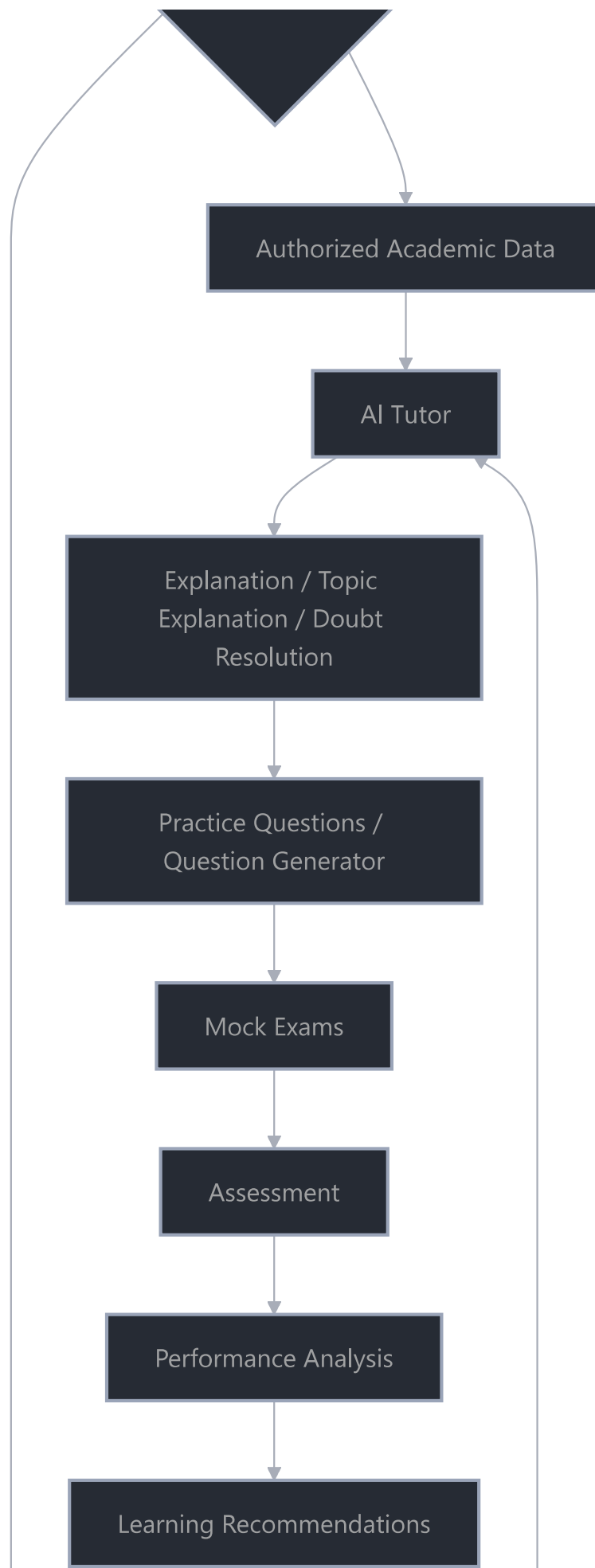
B12. AI Attendance — Detailed Pipeline

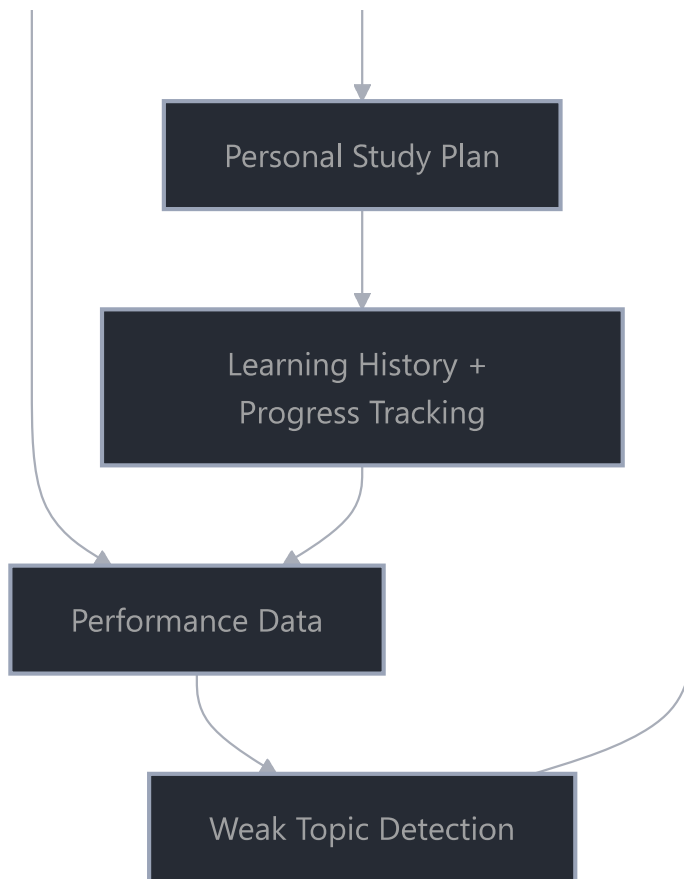




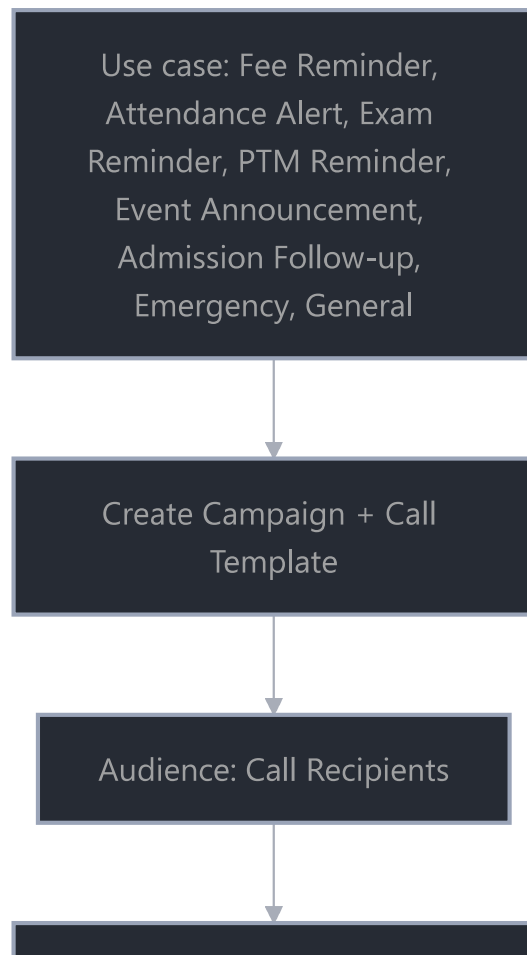
B13. AI Tutor — Learning Loop and Data Scope

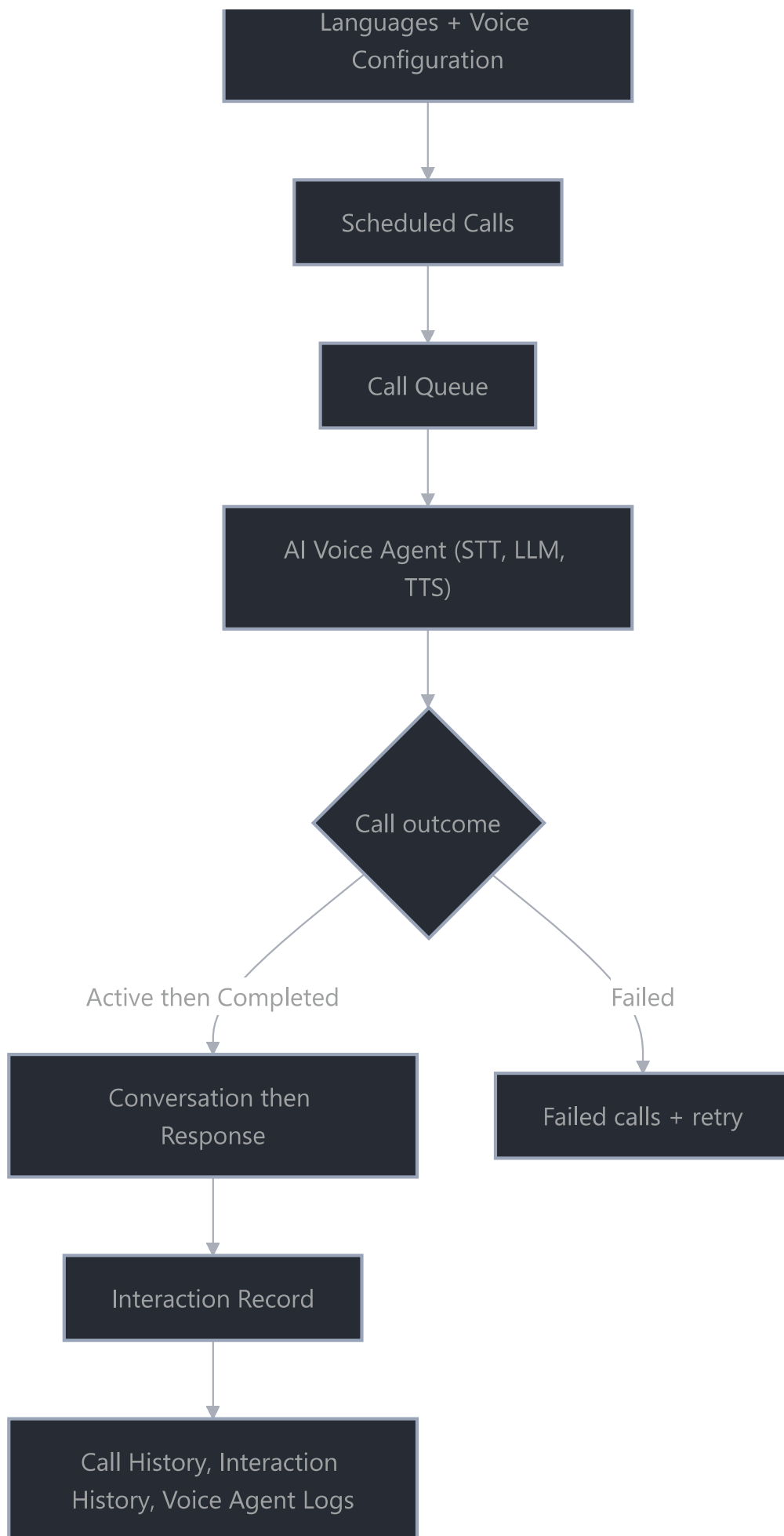




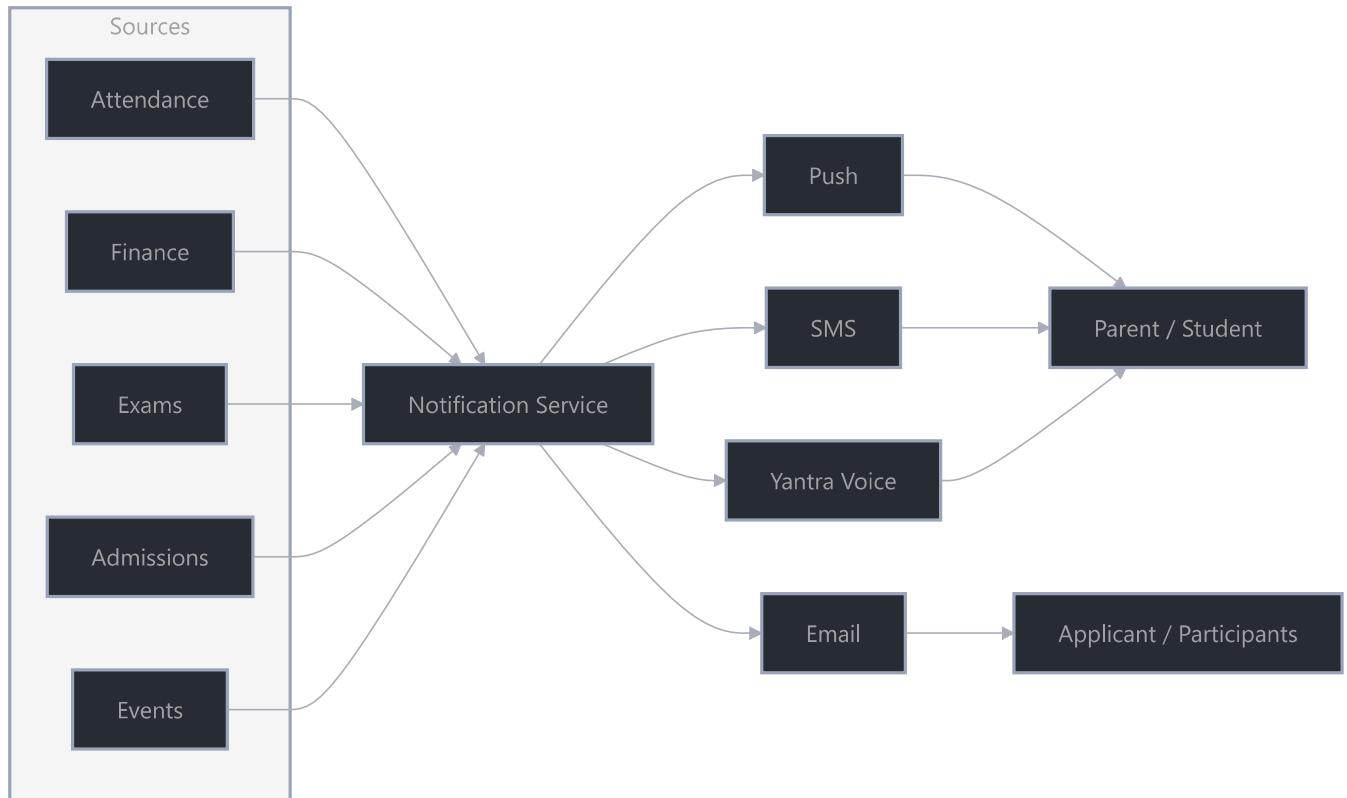


B14. Voice Agent — Campaign Lifecycle

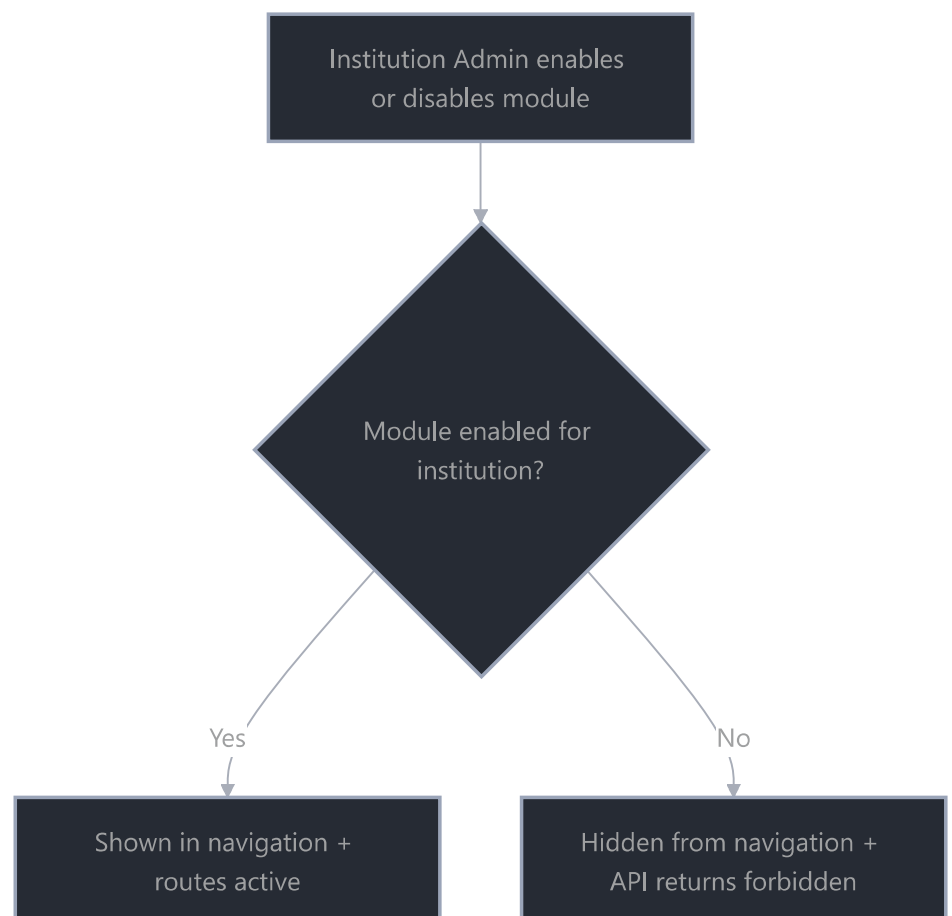


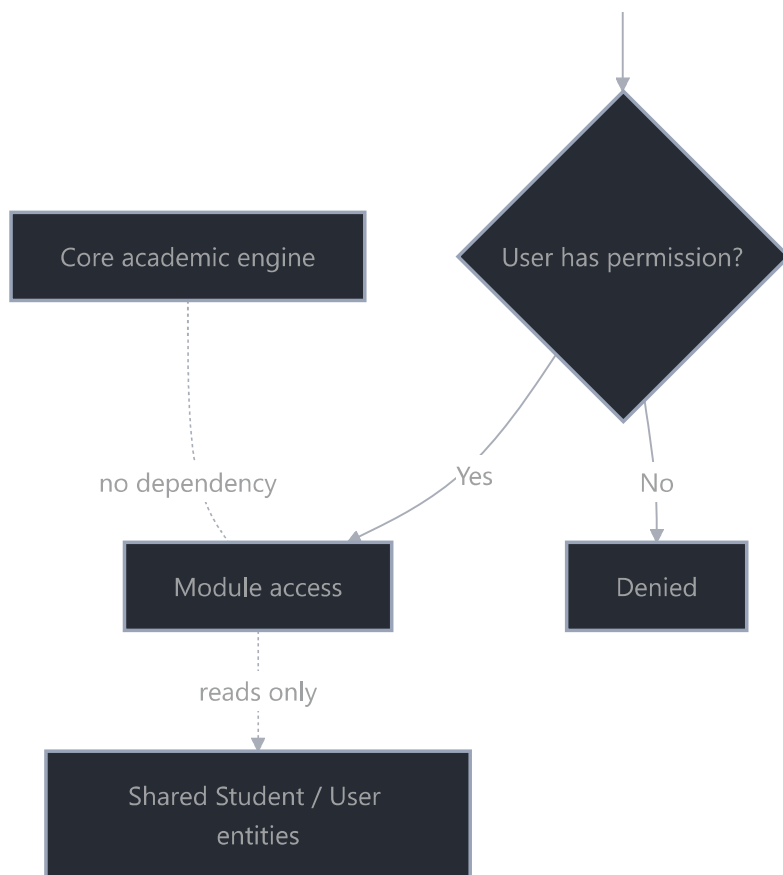


B15. Notification Routing (shared service)

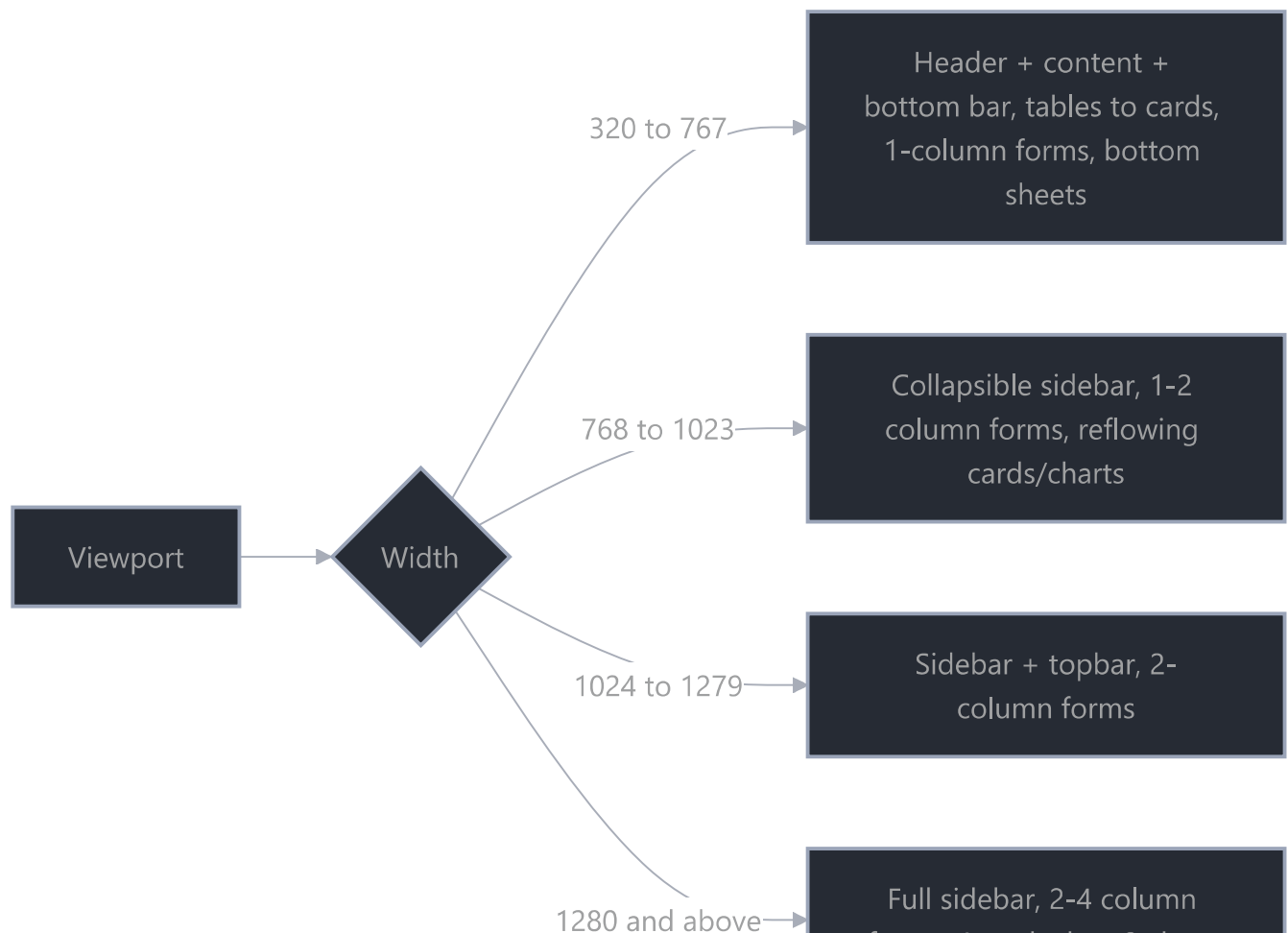


B16. Optional Module Toggle and Isolation

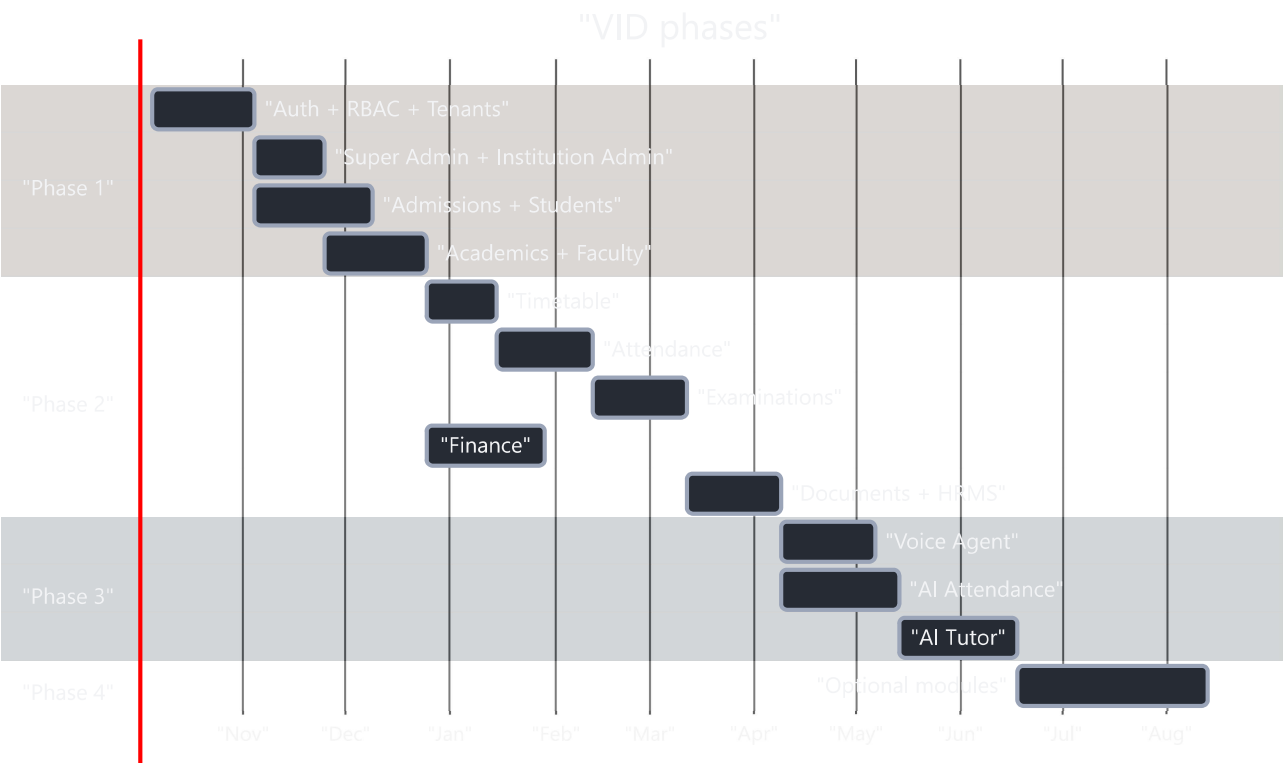




B17. Responsive Layout Behavior



B18. Delivery Roadmap (Gantt)



Dates are illustrative scheduling placeholders; only the phase order comes from the PDF.